
Canadian nutrient file:

users' guide 2026



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To obtain additional information, please contact:

Health Canada
Address Locator 0900C2
Ottawa, ON K1A 0K9
Tel.: 613-957-2991
Toll free: 1-866-225-0709
Fax: 613-941-5366
TTY: 1-800-465-7735
E-mail: publications-publications@hc-sc.gc.ca

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About the Canadian Nutrient File

The Canadian Nutrient File (CNF) is a food composition database containing average nutrient values for foods available in Canada. It includes food names, measure descriptions and background information in both official languages (French and English). Measures follow the metric system (see Appendix A: Metric system equivalents).

The food profiles included in the CNF are regularly updated to ensure they provide accurate and relevant information. These updates help Health Canada carry out food consumption and nutrition surveys, which are important for developing nutrition policies, guidelines, and regulations.

Only foods available in Canada are included, usually as generic foods, except in cases where individual brands are considered unique such as breakfast cereals and chocolate bars.

Each food name, nutrient amount and measure weight conversion record have a “last updated date” field. This field is important for checking how current and relevant the data are for specific uses.

For each food code, the last updated date reflects when a change occurred in any of the fields within the profile.

2026 Edition

This is the fourteenth edition of the Canadian Nutrient File (CNF) providing information on over **5993** foods and up to **173** nutrients.

This standard Canadian resource includes:

- Canadian foods via the Sampling and Nutrient Analysis Program for Canada (SNAP-CAN) [1],
- integration of data from the United States Department of Agriculture (USDA) National Nutrient Database for Standard Reference adapted for the Canadian context [2], and
- data provided by third parties (e.g. commodity groups),

This manual is a technical document meant to guide users of the CNF for:

- appropriate uses of the data;
- background on the sources and quality of the data; and
- information on specific nutrients.

1. Highlight features

Addition of new SNAP-CAN data

The 2026 Canadian Nutrient File (CNF) includes the following new or updated data from the Sampling and Nutrient Analysis Program for Canada (SNAP-CAN) for the following food categories:

- Cheese;
- Chicken strips;
- Cookies;
- Crackers;
- Flatbreads;
- Frozen entrees;
- Frozen pizza;
- Infant snacks;
- Plant-based beverages.

Addition of alternate food description

An alternate food description field has been added to facilitate the search for a particular food in the database. This field can include key words, common names for the food or brand names. For example:

Food Code: 2855

Food Description: Carbonated drinks, ginger ale

Alternate Food Description: pop, soda, soft drink, ginger beer, schweppes, gingerale, ginger ale, Canada Dry

Addition and removal of nutrient codes

The list of published nutrients was reviewed for this edition. Added nutrients can be found in Table 1 while removed nutrients are shown in Table 2. The complete list of nutrients for the 2026 edition can be found in Appendix B: Nutrient count.

Table 1: Added nutrients for 2026 edition

Nutrient Code	Nutrient Symbol	Nutrient Name
326	D3-ug	Vitamin D3, Cholecalciferol
329	25(OH)D3	25-hydroxycholecalciferol
330	25(OH)D2	25-hydroxyergocalciferol
428	MK4	Vitamin K (Phylloquinone)
429	VITK1D	Vitamin K (Dihydrophylloquinone)
665	18:2t	PUFA, 18:2t Not Further Defined, Linoleic, Octadecadienoic
884	18:1 10c	MUFA, 18:1 10c
885	18:1 11c	MUFA, 18:1 11c
886	18:1 12c	MUFA, 18:1 12c
888	18:1 13c	MUFA, 18:1 13c
891	18:1 14c	MUFA, 18:1 14c
895	18:1 15c	MUFA, 18:1 15c
896	18:1 16c	MUFA, 18:1 16c
897	18:1 11t	MUFA, 18:1 11t
898	18:1 4t	MUFA, 18:1 4t
899	18:1 5t	MUFA, 18:1 5t
904	18:1 6t-8t	MUFA, 18:1 6t-8t
905	18:1 10t	MUFA, 18:1 10t
906	18:1 12t	MUFA, 18:1 12t
907	18:1 13t	MUFA, 18:1 13t+14t+6c-8c
908	18:1 16t	MUFA, 18:1 16t
909	20:1 5c	MUFA, 20:1 5c
910	18:2 9c13c	PUFA, 18:2 9c,13c
911	18:2 9c14c	PUFA, 18:2 9c,14c
912	18:2 9c15c	PUFA, 18:2 9c,15c
913	22:5n-6	PUFA, 22:5n-6
1001	INUL	Fructans (Inulin)

Table 2: Removed nutrients for 2026 edition

Nutrient Code	Nutrient Symbol	Nutrient Name
209	STAR	Starch
245	OXAL	Oxalic Acid
902	TOmega n-3	Fatty acids, polyunsaturated, total omega n-3
903	TOmega n-6	Fatty acids, polyunsaturated, total omega n-6

Change in the database file structure

The CNF relational database files can be downloaded as Excel-compatible CSV delimited files available on the Open Government Portal: <https://open.canada.ca>

To view and use these files, you will need a database management software program or a spreadsheet program that can handle large files.

Due to changes in the file structure, files that track changes, additions, or deletions are not provided in this edition. We recommend downloading the entire database for reference.

For more details on the database files, their relationships, and field descriptions, see the document “Canadian nutrient file: database structure and file content description 2026” also available on the Open Government Portal.

2. Strengths and Limitations

The Canadian Nutrient File (CNF) provides average nutrient amounts for generic foods. The nutrient content of, for example, a particular variety of apple or brand of cookie, may differ from the generic average. These averages reflect sources of a food across Canada, but local foods might differ from the national average.

Nutrient values are presented per 100 g of the edible portion of the food.

Missing values

Not every food profile in the CNF contains data for all 173 nutrients. For example, one food profile might include information on 75 nutrients while another might have a complete set of data. Missing values mean data is unavailable, not that the nutrient content is zero. Users need to understand and account for these missing values when using the database. See Appendix B: Nutrient count for the percentage of data available for each nutrients.

Methods of analysis for the nutrients

Most nutrients in the CNF are measured using Association of Official Analytical Chemists (AOAC) methods [3] or other validated procedures. The choice of method depends on the specific component being measured and the type of food. Methods are constantly evolving to improve accuracy, precision and efficiency of results.

Information on the methods used for each nutrient, sourced from the USDA Nutrient Database for Standard Reference, is available in the support files with each Standard Release. You can also find this information on USDA FoodData Central [4].

Bioavailability

Analytical values show the total amount of nutrients in the edible part of the food, including any added during processing. These values do not represent the amounts available to the body, which can be affected by nutrient interactions, physiological mechanisms, nutritional status, and other factors (e.g. cooking).

Applications of the CNF data

The CNF is primarily developed for use in population nutrition surveillance activities.

The CNF is used at Health Canada:

- federal nutrition research,
- risk assessment,
- food regulation, and
- food policy work.

Other uses for this data include:

- dietary assessment,
- recipe development,
- menu planning when ingredients are not specific, and
- early stages of product development to meet nutritional targets.

Limitations for labelling

The CNF provides average values for generic foods. These values come from combining brands of similar products, like major brands of ketchup or different varieties of oranges or similar beef cuts from various producers. Commodity associations might also create generic products with data coming from different producers.

Using generic information from reference databases for labelling purposes is not appropriate because it does not match the product formulation or specific ingredients and processes used in individual products.

3. Sources of food data

The source of each food name and nutrient amount record can be found in the CNF relational files. The database is updated with data from:

- Sampling and Nutrient Analysis Program for Canada (SNAP-CAN),
- United States Department of Agriculture (USDA),
- Other sources.

Sampling and Nutrient Analysis Program for Canada (SNAP-CAN)

The Sampling and Nutrient Analysis Program for Canada (SNAP-CAN) started in 2007 to improve the Canadian Nutrient File (CNF) by incorporating high-quality Canadian data [1]. This program helps Health Canada with nutrition surveillance, research, policies, standards, and regulations by providing nationally representative analytical data for one to three high-priority food categories each year. After choosing the priority food categories, sampling plans are developed to outline which foods will be sampled from the Canadian market. These samples are collected, processed and analyzed for a wide range of nutrients by Health Canada and private laboratories. Results for similar products are combined into generic food profiles with average values calculated based on market share data. Finally, these generic profiles are added into the CNF database.

United States Department of Agriculture (USDA)

In addition to data from SNAP-CAN, the CNF also adapts data from the United States Department of Agriculture (USDA) National Nutrient Database for Standard Reference, up to the Standard Release Legacy [2]. Foods that are not on the Canadian market are excluded. Adjustments for Canadian regulations (e.g. fortification policy) are made when required.

A field called USDA_NDB_Code is a quick reference to find the foods associated with a specific food code in the Food Source file.

Other sources

Data coming from other sources, like traditional (Indigenous) foods, other international databases, recipes, and industry provided data are included in the CNF. However, these make up a very small portion of the database. The complete list of the food and nutrient data sources can be found in the document “Canadian Nutrient File: database structure and file content description 2026” available on the Open Government Portal.

4. Types of nutrient data sources

The Canadian Nutrient File (CNF) uses different types of data to provide nutrient values for foods. These data types reflect how the values were obtained, depending on what information was available. Knowing the source of a nutrient value helps users understand the strengths and limitations. The sections below describe the main types of nutrient data used in the CNF.

Analytical

Analytical data refers to nutrients that have been measured in a laboratory using official methods. This is generally the highest quality of data.

Imputed

Imputed data are nutrient values that are used to fill gaps when analytical or calculated data are not available [5]. Imputed values can be borrowed from:

- similar foods,
- from another form or cooking method of the same food (e.g. values for boiled used for steamed), or
- from other sources (e.g. other food composition databases, scientific literature, food labels).

Calculated

Calculated data refers to nutrient values calculated from analytical or imputed data. They are often used for nutrients that represent the sum of a component such as total saturated fat or to convert data into another unit such as vitamin D in micrograms to international units.

Assumed as zero

Data assumed as zero is used when a nutrient is reasonably presumed to be absent in foods of certain origins. For example, cholesterol for foods of plant origin or total dietary fibre in table sugar.

Calculated from a recipe

Data calculated from a recipe refers to nutrient values that are the result of a calculation based on ingredient proportions. These calculations account for changes in nutrient content and moisture that may occur during cooking.

Provisional

Provisional data refers to nutrient values where there are outstanding questions regarding the sampling procedures and/or methods of analysis, which require further verification.

Information on Nutrients

This section provides an overview of the nutrient data included in the Canadian Nutrient File (CNF). Information is organized into different groupings of nutrients i.e.

1. Proximates
2. Energy
3. Other carbohydrates
4. Minerals
5. Vitamins
6. Amino acids
7. Fatty acids (lipids)
8. Other components

This information is essential for users to interpret nutrient values accurately, ensure data consistency, and apply them effectively in dietary studies, food composition research, and policy development.

1. Proximates

In the context of food composition, “proximate components” refers to the major constituents of a food. In the Canadian Nutrient File (CNF) database, proximate components are:

- moisture (water),
- protein,
- fat (total lipid),
- ash,
- alcohol (when present), and
- carbohydrates.

The sum of these 6 components in 100 g of a food should approach 100 g.

Protein

Proteins are molecules made of long chains of amino acids, molecules that contain nitrogen. In the CNF, protein values are estimated based on the amount of nitrogen in the food, using food-specific conversion factors recommended by Jones (1941) [6]. For foods without a specific factor, the general factor of 6.25 is used.

Protein values for soy products, chocolate, cocoa products, coffee, mushrooms, and yeast are adjusted to account for non-nitrogenous materials.

Table 3: Adjusted protein conversion factors

Product	Adjusted protein conversion factor
Soy products	5.71
Chocolate and cocoa	4.74
Coffee	5.3
Mushrooms	4.38
Yeast	5.7
White flour	5.7
Whole wheat flour	5.83

Protein content is expressed in grams per 100 g (g/100g).

Fat (Total Lipids)

Fat (total lipids) [7] may be analyzed in two ways in the CNF. There is no mechanism to differentiate which method was used in the 2026 release due to technical limitations. The methods are:

1. Triglyceride Equivalents

- Measured using gas chromatography, which analyzes fatty acids.[8]
- Only includes energy-yielding lipids.
- Most accurate for calculating the energy value of food.
- Fatty acids are first measured as fatty acid methyl esters, then converted to free fatty acids, and finally to triglyceride equivalents.

2. Crude Fat

- Measured using gravimetric methods (weighing after solvent extraction).
- Includes all lipid components, even those that do not provide energy (e.g. sterols, terpene alcohols, tocopherols, etc.).
- Useful as a general fat value but may overestimate energy contribution.

Historically fat (total lipids) for SNAP-CAN was measured using the crude fat method. More recently, fat (total lipids) has been measure with one of the two methods for triglycerides.

Fat (total lipids) content is expressed in grams per 100 g (g/100g).

Ash

Ash content refers to the residue left after a food sample is burned or heated to a very high temperature, removing all the water and organic material. Ash is used in the calculation of carbohydrate by difference (see Carbohydrates).

Ash content is expressed in grams per 100 g (g/100g).

Alcohol

Alcohol content refers to the amount of alcohol present in a given food or beverage. When present, alcohol content contributes to the calculation of the food's energy content (kilocalories).

Information on alcohol content is mostly limited to:

- alcoholic beverages,
- a few culinary ingredients, and
- some recipes including these ingredients.

Alcohol is assumed zero in most foods in the CNF.

Alcohol content is expressed in grams per 100 g (g/100g).

Carbohydrates

Carbohydrates are a diverse group of substances with different chemical, physical, and physiological properties. They are primarily grouped as:

- sugars;
- oligosaccharides; and
- polysaccharides.

They are mainly classified based on their chemistry, including the type of individual monomers, degree of polymerization (DP), and type of linkage (α or β). It is important to note that the CNF does not provide information for all the individual carbohydrate components listed in Appendix D: Class of carbohydrates

The carbohydrate value is determined as the difference between 100 and the sum of the remaining proximate components.

$$\text{Carbohydrate} = 100 - (\text{Moisture} + \text{Protein} + \text{Total Lipid} + \text{Ash} + \text{Alcohol [if present]})$$

This method of calculation is also known as carbohydrate-by-difference. The determination of total carbohydrate values by this method of calculation includes total dietary fibre.

Because the analyses of total dietary fibre, total sugars, and starch (including maltodextrin) are done separately and have some variability in measurement, the sum of these carbohydrate fractions might not exactly equal the carbohydrate-by-difference value.

Carbohydrate content is expressed in grams per 100 g (g/100g).

2. Energy

Food energy is expressed in both kilocalories (kcal) and kilojoules (kJ).

$$1 \text{ kcal} = 4.184 \text{ kJ}$$

Caloric values are calculated using the Atwater system, which determines energy values. Most foods in the CNF use the Atwater general factors [9] to calculate calories.

Nutrient	Caloric Value
Protein	4 kcal/g
Fat	9 kcal/g
Carbohydrates	4 kcal/g
Alcohol (when present)	7 kcal/g

However, some caloric values are based on the specific Atwater factors.

3. Other Carbohydrates

Total Dietary Fibre

Total dietary fibre (TDF) is made of complex materials that are difficult to separate from other food components, particularly starch. Several approved methods are available to measure TDF content.

Total dietary fibre is assumed zero in many foods after reviewing literature and/or consultation with scientific experts (nutrient source code 12).

Total dietary fibre content is expressed in grams per 100 g (g/100g).

Fructans

Many foods contain small amounts of fructans. Generally, only foods with a significant amount of added fructans are analysed for SNAP-CAN. The value from the analysis is then included in the total dietary fibre value. Inulin is the most commonly added source.

Sugars and Total Sugars

Monosaccharides (glucose, fructose and galactose) and disaccharides (sucrose, lactose and maltose) are commonly referred to as sugars.

Total sugars is defined as the sum of the monosaccharides and disaccharides. It does not include sugar alcohols, which are found in small amounts in some fruit and sometimes added to foods as sweeteners. It also does not include maltodextrins, which are easily digested and absorbed like other α -glucans and often added as sweeteners, fat substitutes, and to change the texture of foods.

$$\text{Total sugars} = \text{Glucose} + \text{Fructose} + \text{Galactose} + \text{Sucrose} + \text{Lactose} + \text{Maltose}$$

Values for individual sugars and total sugars content are expressed in grams per 100 g (g/100g).

4. Minerals

Minerals in food are essential inorganic elements that the body needs for various functions. They must be obtained from the diet since the body can't produce them. The CNF includes data on the following minerals:

- calcium,
- iron,
- magnesium,
- phosphorus,
- potassium,
- sodium,
- zinc,
- copper,
- manganese, and
- selenium.

Minerals are expressed in milligrams per 100 g (mg/100g) except for selenium which is expressed in micrograms per 100 g (µg/100g).

5. Vitamins

This section provides an overview of some of the vitamins included in the Canadian Nutrient File (CNF). The following vitamins are described with more details on forms found in foods below:

- Vitamin A,
- Vitamin D,
- Vitamin E,
- Niacin,
- Folate and folic acid.

Vitamin A

Retinol is found in foods from animal sources. Provitamin A carotenoids are plant pigments that turn into vitamin A in the body. The body can't use these forms until they are converted into retinol.

Carotenoids available in the CNF include:

- retinol,
- lycopene,
- α -carotene (alpha-carotene),
- β -carotene (beta-carotene),
- β -cryptoxanthin (beta-cryptoxanthin), and
- lutein & zeaxanthin (combined)

They are all expressed in micrograms per 100 g ($\mu\text{g}/100\text{g}$).

Vitamin A activity

Some carotenoids are precursors of vitamin A and their values are used in the Retinol Activity Equivalents (RAE) calculation. Vitamin A activity in foods is measured by converting provitamin A carotenoids to RAE and then combining them with the retinol content. There are different ways to measure this total activity, and no single analysis method has been adopted.

Total vitamin A and carotenoids are expressed in micrograms of Retinol Activity Equivalents (RAE) [10]:

$$1 \text{ RAE} = 1 \mu\text{g retinol} + 1 \mu\text{g } \beta\text{-carotene}/12 + 1 \mu\text{g } \alpha\text{-carotene}/24 + 1 \mu\text{g } \beta\text{-cryptoxanthin}/24$$

Vitamin D

Very few foods naturally contain vitamin D. As such, fortified foods represent the majority of the vitamin D consumed by Canadians [11]. Cholecalciferol, or vitamin D3, is found naturally in animal products. Ergocalciferol, or vitamin D2, is naturally found in some plants. Both forms can be found in dietary supplements and fortified foods [12].

The CNF database reports the total amount of D2 and D3 and in some cases the individual D2 and D3 values. Some data on the vitamin D metabolites called 25 hydroxy-D3 [13] and 25 hydroxy-D2 are now available for SNAP-CAN data.

Vitamin D is expressed in micrograms (μg) and International Units (IU) per 100g.

$$1 \mu\text{g} = 40 \text{ IU vitamin D}$$

Vitamin E

There are eight different forms of vitamin E [14, 15], but they are not all equivalent in activity. As such, vitamin E is shown as α -tocopherol (alpha-tocopherol) in milligrams (mg) in the CNF. While beta, gamma and delta tocopherol do not contribute to the vitamin E value, they are included in the CNF database when values are available.

Alpha-tocopherol, beta-tocopherol, gamma-tocopherol and delta-tocopherol are expressed in milligrams per 100 g (mg/100g).

Niacin

Niacin is expressed both in terms of milligrams (mg) of preformed niacinamide present in the food as well as niacin equivalents (NE) which can be formed from tryptophan. There are 2 methods of calculating niacin equivalents [10].

Method 1: If data for preformed niacin (in milligrams) and tryptophan (in grams) are available niacin activity is calculated using the following formula:

$$NE = (\text{tryptophan} \times 1000/60) + \text{preformed niacin}$$

Method 2: If data for tryptophan (in grams) is not available but preformed niacin (in milligrams) and protein (in grams) are present then niacin activity is imputed to be 1.1% of total protein using the following formula

$$NE = ([0.011 \times \text{protein}] \times 1000/60) + \text{preformed niacin}$$

Folate and folic acid

Folate occurs in foods as:

- naturally occurring folate (food folate)
- added synthetic form (folic acid)

Folic acid is more bioavailable than food folate [16, 17]. The CNF includes data on the following forms:

- food folate (naturally occurring) in micrograms per 100 g ($\mu\text{g}/100 \text{ g}$)
- folic acid in micrograms per 100 g ($\mu\text{g}/100 \text{ g}$)

- total folacin - the sum of folic acid and food folate, not accounting for activity, in micrograms per 100 g ($\mu\text{g}/100\text{ g}$)
- Dietary Folate Equivalents (DFE) in micrograms per 100 g ($\mu\text{g}/100\text{ g}$)

$$1 \text{ DFE} = (\mu\text{g folic acid} \times 1.7) + \mu\text{g food folate}$$

6. Amino acids

Amino acids form the building blocks of the total protein content in foods. The sum of amino acids will be close, but not identical to total protein. They are analyzed by a different method of analysis which can result in variations

For SNAP-CAN data, variation may also result from the analysis method : total protein was measured in individual food products, while amino acids were measured in composited samples.

Amino acid values are expressed in grams per 100 g ($\text{g}/100\text{ g}$) in the CNF.

7. Fatty Acids (Lipids)

Nomenclature

Fatty acids are named using different systems. They are referred by isomer most commonly. The most specific way to describe these isomers uses a shorthand system with numbers and letters:

- **Carbon Atoms and Double Bonds:** The first number (before the colon) indicates the number of carbon atoms. The second number (after the colon) shows the number of double bonds in the chain.
- **Bond Types:** The letters c, t, or i indicate if the bond is cis, trans, or a mix of both.

For example, 18:2 t,t means a fatty acid with 18 carbon atoms, 2 double bonds, and both bonds are trans.

Undifferentiated Fatty Acids

Current methods used in SNAP-CAN data can identify cis and trans isomers separately.

When the word “undifferentiated” or “undiff” is part of the fatty acid name, it means the proportions of cis and trans are unknown because the values were added before they were analyzed separately. For these foods, “undifferentiated” or “undiff” fields are the total of all isomers (calculated field). For example:

18:2undiff is the sum of 18:2ccn-6, 18:2t,t , 18:2i and 18:2cla

Omega-3 and Omega-6

Omega-3 and omega-6 isomers are shown as n-3 and n-6. The n-number shows the position of the first double bond from the end of the carbon chain.

For example, 18:2 c,c n-6 means a fatty acid with 18 carbon atoms, 2 double bonds, the first double is in the omega 6 position and both double bonds are cis.

Fatty Acid Totals

Appendix C: List of fatty acids lists the fatty acids included in the different totals:

- saturated fatty acids (TSAT)
- monounsaturated fatty acids (MUFA),
- polyunsaturated fatty acids (PUFA),
- trans fatty acids (TRFA)
- trans mono fatty acids (TRMO)
- trans poly fatty acids (TRPO)

Individual fatty acids and total fat are measured using different methods, which is why the sum of individual fatty acids rarely matches the total fat (total lipids) value exactly.

Expression of Fatty Acids Values

The values and fatty acid totals shown represent the free fatty acid content [18] in grams per 100 g (g/100 g) and do not represent triglycerides.

8. Other components

Cholesterol

Cholesterol is present only in foods of animal origin or containing foods of animal origin. For foods of plant origin, the value for cholesterol is assumed to be zero (nutrient source code 12).

Cholesterol is expressed in milligrams per 100 g (mg/100 g).

Glossary

Table : List of acronyms

Acronym	Definition
AOAC	Association of Official Analytical Chemists
CLA	Conjugated Linoleic Acid
CNF	Canadian Nutrient File
DFE	Dietary Folate Equivalent
DP	Degree of polymerization
g	Gram
INFOODS	International Network of Food Data Systems
IU	International Unit
kcal	Kilocalorie
kJ	Kilojoule
mg	Milligram
MUFA	Monounsaturated Fatty Acids
NE	Niacin Equivalent
PUFA	Polyunsaturated Fatty Acids
RAE	Retinol Activity Equivalent
SNAP-CAN	Sampling and Nutrient Analysis Program for Canada
TDF	Total Dietary Fibre
TRFA	Trans Fatty Acids
TRMO	Trans Mono Fatty Acids
TRPO	Trans Poly Fatty Acids
TSAT	Total Saturated Fatty Acids
µg	Microgram
USDA	United States Department of Agriculture

Appendix A: Metric system equivalents

Table: Metric system equivalents for units of measure

	Metric System Equivalents	Canadian Metric household measure		US and Imperial measures
Volume	4.9 ml	5 ml	n/a	1 teaspoon
	14.8 ml	15.0 ml	n/a	1 tablespoon
	29.57 ml	n/a	n/a	1 fluid ounce (US)
	28.41 ml	n/a	n/a	1 fluid ounce (Imperial)
	236.6 ml	250 ml	n/a	1 cup (8 US fluid ounces)
	473.2 ml	n/a	n/a	1 pint (16 US fluid ounces)
	568.3 ml	500 ml	n/a	1 pint (20 Imperial fluid ounces)
	946.4 ml	n/a	n/a	1 quart (32 US fluid ounces)
	1136.5 ml	1 L	n/a	1 quart (40 Imperial fluid ounces)
	3786 ml	n/a	n/a	1 gallon (128 US fluid ounces)
	4546 ml	4 L	n/a	1 gallon (160 Imperial fluid ounces)
	16.39 ml	2.54 cm cube	15.63 ml	1 cubic inch
Length	2.54 cm, 25.40 mm	n/a	n/a	1 inch
Weight	28.35 g	n/a	n/a	1 ounce
	453.6 g	n/a	n/a	1 pound
	140 g	250 ml	148 g	1 cup (poultry and cooked meats chopped and diced)
	110 g	250 ml	116 g	1 cup (poultry and cooked meats ground)
Energy	4.184 kJ	n/a	n/a	1 Calorie (1 kcal)

Appendix B: Nutrient count

Percentage of foods containing the indicated nutrient for which a value is available. For each nutrient, the percentage is calculated as the number of foods containing an entry for the nutrient divided by the total number of foods in the database.

Table: List nutrients

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
203	PROT	gram	Protein	2	5993	100.00%
204	FAT	gram	Fat (Total lipids)	2	5993	100.00%
205	CARB	gram	Carbohydrate, Total (by difference)	2	5993	100.00%
207	ASH	gram	Ash, Total	2	5993	100.00%
208	KCAL	kilocalorie	Energy (Kilocalories)	0	5993	100.00%
210	SUCR	gram	Sucrose	2	3122	52.09%
211	GLUC	gram	Glucose	2	3118	52.03%
212	FRUC	gram	Fructose	2	3115	51.98%
213	LACT	gram	Lactose	2	3094	51.63%
214	MALT	gram	Maltose	2	3074	51.29%
221	ALCO	gram	Alcohol	2	5770	96.28%
255	H2O	gram	Moisture	2	5993	100.00%
260	MANN	gram	Mannitol	2	1457	24.31%
261	SORB	gram	Sorbitol	2	1464	24.43%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
262	CAFF	milligram	Caffeine	0	5712	95.31%
263	THBR	milligram	Theobromine	0	5691	94.96%
268	KJ	kilojoule	Energy (Kilojoules)	0	5993	100.00%
269	TSUG	gram	Sugars, Total	2	5029	83.91%
287	GAL	gram	Galactose	2	3046	50.83%
291	TDF	gram	Fibre, Total Dietary	1	5774	96.35%
301	CA	milligram	Calcium	0	5954	99.35%
303	FE	milligram	Iron	2	5954	99.35%
304	MG	milligram	Magnesium	0	5794	96.68%
305	P	milligram	Phosphorus	0	5853	97.66%
306	K	milligram	Potassium	0	5835	97.36%
307	NA	milligram	Sodium	0	5950	99.28%
309	ZN	milligram	Zinc	2	5788	96.58%
312	CU	milligram	Copper	3	5740	95.78%
315	MN	milligram	Manganese	3	5403	90.16%
317	SE	microgram	Selenium	1	5286	88.20%
319	RT-ug	microgram	Retinol	0	5354	89.34%
320	RAE	microgram	Retinol Activity Equivalents	0	5664	94.51%
321	BC-ug	microgram	Beta-Carotene	0	5081	84.78%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
322	AC-ug	microgram	Alpha-Carotene	0	3507	58.52%
323	ATMG	milligram	Alpha-Tocopherol	2	4370	72.92%
324	D3+D2-IU	International Unit	Vitamin D (International Units)	0	5277	88.05%
325	D2-ug	microgram	Vitamin D2, Ergocalciferol	1	644	10.75%
326 (NEW)	D3-ug	microgram	Vitamin D3, Cholecalciferol	1	202	3.37%
328	D3+D2-ug	microgram	Vitamin D (D2+D3)	1	5277	88.05%
329 (NEW)	25(OH)D3	microgram	25-hydroxycholecalciferol	1	0	0%
330 (NEW)	25(OH)D2	microgram	25-hydroxyergocalciferol	1	1	0.02%
334	CRYPX	microgram	Beta-Cryptoxanthin	0	3519	58.72%
337	LYCPN	microgram	Lycopene	0	3525	58.82%
338	LUT+ZEA	microgram	Lutein and Zeaxanthin	0	3504	58.47%
341	BTMG	milligram	Beta-Tocopherol	2	868	14.48%
342	GTMG	milligram	Gamma-Tocopherol	2	874	14.58%
343	DTMG	milligram	Delta-Tocopherol	2	868	14.48%
401	VITC	milligram	Vitamin C	1	5786	96.55%
404	THIA	milligram	Thiamine	3	5725	95.53%
405	RIBO	milligram	Riboflavin	3	5741	95.80%
406	N-MG	milligram	Niacin (Nicotinic Acid) Preformed	3	5757	96.06%
409	N-NE	Niacin Equivalent	Total Niacin Equivalent	3	5759	96.10%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
410	PANT	milligram	Pantothenic Acid	3	4830	80.59%
415	B6	milligram	Vitamin B-6	3	5616	93.71%
416	BIOT	microgram	Biotin	1	116	1.94%
417	FOLA	microgram	Total Folacin	0	5581	93.13%
418	B12	microgram	Vitamin B-12	2	5594	93.34%
421	CHOLN	milligram	Choline, Total	1	3134	52.29%
428 (NEW)	MK4	microgram	Vitamin K (Menaquinone-4)	1	151	2.52%
429 (NEW)	VITK1D	microgram	Vitamin K (Dihydrophyloquinone)	1	70	1.17%
430	VITK	microgram	Vitamin K (Phylloquinone)	1	3460	57.73%
431	FOAC	microgram	Folic Acid	0	5825	97.20%
432	FOLN	microgram	Naturally Occurring Folate	0	5506	91.87%
435	DFE	microgram	Dietary Folate Equivalents	0	5504	91.84%
454	BETN	milligram	Betaine	1	1138	18.99%
501	TRP	gram	Tryptophan	3	4022	67.11%
502	THR	gram	Threonine	3	4077	68.03%
503	ISO	gram	Isoleucine	3	4081	68.10%
504	LEU	gram	Leucine	3	4077	68.03%
505	LYS	gram	Lysine	3	4095	68.33%
506	MET	gram	Methionine	3	4092	68.28%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
507	CYS	gram	Cystine	3	4017	67.03%
508	PHE	gram	Phenylalanine	3	4077	68.03%
509	TYR	gram	Tyrosine	3	4048	67.55%
510	VAL	gram	Valine	3	4081	68.10%
511	ARG	gram	Arginine	3	4068	67.88%
512	HIS	gram	Histidine	3	4075	68.00%
513	ALA	gram	Alanine	3	4023	67.13%
514	ASP	gram	Aspartic Acid	3	4009	66.89%
515	GLU	gram	Glutamic Acid	3	4026	67.18%
516	GLY	gram	Glycine	3	4024	67.15%
517	PRO	gram	Proline	3	4016	67.01%
518	SER	gram	Serine	3	4015	66.99%
521	HYP	gram	Hydroxyproline	3	632	10.55%
550	ASPA	milligram	Aspartame	0	88	1.47%
573	ATMG-A	milligram	Alpha-Tocopherol, Added	2	705	11.76%
578	B12-A	microgram	Vitamin B12, Added	2	731	12.20%
601	CHOL	milligram	Cholesterol	2	5744	95.85%
605	TRFA	gram	Fatty Acids, Trans, Total	3	5068	84.57%
606	TSAT	gram	Fatty Acids, Saturated, Total	3	5767	96.23%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
607	4:0	gram	Fatty Acids, Saturated, 4:0, Butanoic	3	4232	70.62%
608	6:0	gram	Fatty Acids, Saturated, 6:0, Hexanoic	3	4253	70.97%
609	8:0	gram	Fatty Acids, Saturated, 8:0, Octanoic	3	4390	73.25%
610	10:0	gram	Fatty Acids, Saturated, 10:0, Decanoic	3	4678	78.06%
611	12:0	gram	Fatty Acids, Saturated, 12:0, Dodecanoic	3	4837	80.71%
612	14:0	gram	Fatty Acids, Saturated, 14:0, Tetradecanoic	3	5233	87.32%
613	16:0	gram	Fatty Acids, Saturated, 16:0, Hexadecanoic	3	5420	90.44%
614	18:0	gram	Fatty Acids, Saturated, 18:0, Octadecanoic	3	5407	90.22%
615	20:0	gram	Fatty Acids, Saturated, 20:0, Eicosanoic	3	2426	40.48%
617	18:1undiff	gram	MUFA, 18:1undifferentiated, Octadecenoic	3	5440	90.77%
618	18:2undiff	gram	PUFA, 18:2undifferentiated, Linoleic, Octadecadienoic	3	5458	91.07%
619	18:3undiff	gram	PUFA, 18:3undifferentiated, Linolenic, Octadecatrienoic	3	5370	89.60%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
620	20:4undiff	gram	PUFA, 20:4, Arachidonic, Eicosatetraenoic	3	4851	80.94%
621	22:6n-3DHA	gram	PUFA, 22:6 n-3, Docosahexaenoic (DHA)	3	5848	97.58%
624	22:0	gram	Fatty Acids, Saturated, 22:0, Docosanoic	3	2385	39.80%
625	14:1	gram	MUFA, 14:1, Tetradecenoic	3	2404	40.11%
626	16:1undiff	gram	MUFA, 16:1undifferentiated, Hexadecenoic	3	5208	86.90%
627	18:4	gram	PUFA, 18:4, Octadecatetraenoic	3	4291	71.60%
628	20:1	gram	MUFA, 20:1, Eicosenoic	3	4307	71.87%
629	20:5n-3EPA	gram	PUFA, 20:5 n-3, Eicosapentaenoic (EPA)	3	4765	79.51%
630	22:1undiff	gram	MUFA, 22:1undifferentiated, Docosenoic	3	4539	75.74%
631	22:5n-3DPA	gram	PUFA, 22:5 n-3, Docosapentaenoic (DPA)	3	5834	97.35%
636	TPST	milligram	Total Plant Sterol	0	719	12.00%
638	STIG	milligram	Stigmasterol	0	535	8.93%
639	CAMSTR	milligram	Campesterol	0	285	4.76%
641	SITSTR	milligram	Beta-Sitosterol	0	496	8.28%
645	MUFA	gram	Fatty Acids, Monounsaturated, Total	3	5679	94.76%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
646	PUFA	gram	Fatty Acids, Polyunsaturated, Total	3	5677	94.73%
652	15:0	gram	Fatty Acids, Saturated, 15:0, Pentadecanoic	3	2297	38.33%
653	17:0	gram	Fatty Acids, Saturated, 17:0, Heptadecanoic	3	2344	39.11%
654	24:0	gram	Fatty Acids, Saturated, 24:0, Tetracosanoic	3	2119	35.36%
662	16:1t	gram	MUFA, 16:1t, Hexadecenoic	3	2101	35.06%
663	18:1t	gram	MUFA, 18:1t, Octadecenoic	3	1942	32.40%
664	22:1t	gram	MUFA, 22:1t, Docosenoic	3	3053	50.94%
665 (NEW)	18:2t	gram	PUFA, 18:2t Not Further Defined, Linoleic, Octadecadienoic	3	338	5.64%
666	18:2i	gram	PUFA, 18:2i, Linoleic, Octadecadienoic	3	1718	28.67%
669	18:2t,t	gram	PUFA, 18:2t,t, Octadecadienenoic	3	1384	23.09%
670	18:2cla	gram	PUFA, Conjugated, 18:2 cla, Linoleic, Octadecadienoic	3	1731	28.88%
671	24:1c	gram	MUFA, 24:1c, Tetracosenoic	3	1875	31.29%
672	20:2cc	gram	PUFA, 20:2 c,c, Eicosadienoic	3	2218	37.01%
673	16:1c	gram	MUFA, 16:1c, Hexadecenoic	3	2137	35.66%
674	18:1c	gram	MUFA, 18:1c, Octadecenoic	3	1928	32.17%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
675	18:2ccn-6	gram	PUFA, 18:2 c,c n-6, Linoleic Octadecadienoic	3	2773	46.27%
676	22:1c	gram	MUFA, 22:1c, Docosenoic	3	3125	52.14%
685	18:3cccn-6	gram	PUFA, 18:3 c,c,c n-6, g-linolenic, Octadecatrienoic	3	5597	93.39%
687	17:1	gram	MUFA, 17:1, Heptadecenoic	3	2166	36.14%
689	20:3undiff	gram	PUFA, 20:3 undifferentiated, Eicosatrienoic	3	3609	60.22%
693	TRMO	gram	Fatty Acids, Total trans-monoenoic	3	1790	29.87%
695	TRPO	gram	Fatty Acids, Total trans-polyenoic	3	1711	28.55%
696	13:0	gram	Fatty Acids, Saturated 13:0, Tridecanoic	3	661	11.03%
697	15:1	gram	MUFA, 15:1, Pentadecenoic	3	2021	33.72%
851	18:3cccn-3	gram	PUFA, 18:3 c,c,c n-3, Linolenic, Octadecatrienoic	3	4987	83.21%
852	20:3n-3	gram	PUFA, 20:3 n-3, Eicosatrienoic	3	5394	90.01%
853	20:3n-6	gram	PUFA, 20:3 n-6	3	5376	89.70%
855	20:4n-6	gram	PUFA, 20:4 n-6, Arachidonic	3	3491	58.25%
856	18:3i	gram	PUFA, 18:3i, Linolenic, Octadecatrienoic	3	1657	27.65%
857	21:5	gram	PUFA, 21:5	3	1266	21.12%
858	22:4n-6	gram	PUFA, 22:4 n-6, Docosatetraenoic	3	1833	30.59%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
859	24:1undiff	gram	MUFA, 24:1undifferentiated, Tetracosenoic	3	1153	19.24%
860	12:1	gram	MUFA, 12:1, Lauroleic	3	582	9.71%
861	22:3	gram	PUFA, 22:3	3	1084	18.09%
862	22:2	gram	PUFA, 22:2, Docosadienoic	3	1311	21.88%
884 (NEW)	18:1 10c	gram	MUFA, 18:1 10c	3	358	5.97%
885 (NEW)	18:1 11c	gram	MUFA, 18:1 11c	3	358	5.97%
886 (NEW)	18:1 12c	gram	MUFA, 18:1 12c	3	358	5.97%
888 (NEW)	18:1 13c	gram	MUFA, 18:1 13c	3	358	5.97%
891 (NEW)	18:1 14c	gram	MUFA, 18:1 14c	3	358	5.97%
895 (NEW)	18:1 15c	gram	MUFA, 18:1 15c	3	358	5.97%
896 (NEW)	18:1 16c	gram	MUFA, 18:1 16c	3	358	5.97%
897 (NEW)	18:1 11t	gram	MUFA, 18:1 11t	3	358	5.97%
898 (NEW)	18:1 4t	gram	MUFA, 18:1 4t	3	358	5.97%
899 (NEW)	18:1 5t	gram	MUFA, 18:1 5t	3	358	5.97%
904 (NEW)	18:1 6t-8t	gram	MUFA, 18:1 6t-8t	3	358	5.97%
905 (NEW)	18:1 10t	gram	MUFA, 18:1 10t	3	358	5.97%
906 (NEW)	18:1 12t	gram	MUFA, 18:1 12t	3	358	5.97%
907 (NEW)	18:1 13t	gram	MUFA, 18:1 13t+14t+6c-8c	3	355	5.92%
908 (NEW)	18:1 16t	gram	MUFA, 18:1 16t	3	358	5.97%

Nutrient Code	Nutrient Symbol	Unit	Nutrient Name	# Decimal Places	#Foods	Percent
909 (NEW)	20:1 5c	gram	MUFA, 20:1 5c	3	358	5.97%
910 (NEW)	18:2 9c13c	gram	PUFA, 18:2 9c,13c	3	358	5.97%
911 (NEW)	18:2 9c14c	gram	PUFA, 18:2 9c,14c	3	358	5.97%
912 (NEW)	18:2 9c15c	gram	PUFA, 18:2 9c,15c	3	358	5.97%
913 (NEW)	22:5n-6	gram	PUFA, 22:5n-6	3	358	5.97%
1001 (NEW)	INUL	gram	(Fructans) Inulin	1	53	0.88%

Appendix C: List of fatty acids contributing to different totals

Table: List of fatty acids in the Canadian Nutrient File and their contribution to the different totals*

Nutrient Code	Nutrient Symbol	Fatty Acids	Nutrient Name	Common Name of Typical Isomer	Total Fatty Acids Category
607	4:0	4:0	Butanoic	Butyric	TSAT
608	6:0	6:0	Hexanoic	Caproic	TSAT
609	8:0	8:0	Octanoic	Caprylic	TSAT
610	10:0	10:0	Decanoic	Capric	TSAT
611	12:0	12:0	Dodecanoic	Lauric	TSAT
696	13:0	13:0	Tridecanoic	N/A	TSAT
612	14:0	14:0	Tetradecanoic	Myristic	TSAT
652	15:0	15:0	Pentadecanoic	Pentadecylic	TSAT
613	16:0	16:0	Hexadecanoic	Palmitic	TSAT
653	17:0	17:0	Heptadecanoic	Margaric	TSAT
614	18:0	18:0	Octadecanoic	Stearic	TSAT
615	20:0	20:0	Eicosanoic	Arachidic	TSAT
624	22:0	22:0	Docosanoic	Behenic	TSAT
654	24:0	24:0	Tetracosanoic	Lignoceric	TSAT
860	12:1	12:1	Lauroleic	N/A	MUFA
625	14:1	14:1	Tetradecenoic	Myristoleic	MUFA
697	15:1	15:1	Pentadecenoic	N/A	MUFA
626	16:1 undiff	16:1 undifferentiated	Hexadecenoic	Palmitoleic	MUFA
687	17:1	17:1	Heptadecenoic	N/A	MUFA

Nutrient Code	Nutrient Symbol	Fatty Acids	Nutrient Name	Common Name of Typical Isomer	Total Fatty Acids Category
617	18:1 undiff	18:1 undifferentiated	Octadecenoic	Oleic	MUFA
628	20:1	20:1	Eicosenoic	Gadoleic	MUFA
630	22:1 undiff	22:1 undifferentiated	Docosenoic	Erucic	MUFA
859	24:1 undiff	24:1 undifferentiated	Tetracosenoic	Nervonic	MUFA
618	18:2 undiff	18:2 undifferentiated	Linoleic, octadecadienoic	Linoleic	PUFA
619	18:3 undiff	18:3 undifferentiated	Linolenic, octadecatrienoic	Linolenic	PUFA
627	18:4	18:4	Octadecatetraenoic	Parinaric	PUFA
672	20:2cc	20:2cc	Eicosadienoic	N/A	PUFA
689	20:3 undiff	20:3 undifferentiated	Eicosatrienoic	N/A	PUFA
620	20:4 undiff	20:4	Arachidonic, Eicosatetraenoic	Arachidonic	PUFA
629	20:5n-3 (EPA)	20:5n-3 (EPA)	Eicosapentaenoic (EPA)	Timnodonic	PUFA
857	21:5	21:5	N/A	N/A	PUFA
862	22:2	22:2	Docosadienoic	N/A	PUFA
861	22:3	22:3	N/A	N/A	PUFA
858	22:4n-6	22:4n-6	Docosatetraenoic	N/A	PUFA
631	22:5n-3 (DPA)	22:5n-3 (DPA)	Docosapentaenoic (DPA)	Clupanodonic	PUFA
913 (NEW)	22:5n-6	22:5n-6	N/A	N/A	PUFA

Nutrient Code	Nutrient Symbol	Fatty Acids	Nutrient Name	Common Name of Typical Isomer	Total Fatty Acids Category
621	22:6n-3 (DHA)	22:6n-3 (DHA)	Docosahexaenoic (DHA)	N/A	PUFA
662	16:1t	16:1t	Hexadecenoic	N/A	TRFA; TRMO
663	18:1t	18:1t	Octdecenoic	N/A	TRFA; TRMO
664	22:1t	22:1t	Docosenoic	N/A	TRFA; TRMO
665 (NEW)	18:2t	18:2t	Linoleic, octadecadienoic	N/A	TRFA; TRPO
666	18:2i	18:2i	Linoleic, octadecadienoic	N/A	TRFA; TRPO
856	18:3i	18:3i	Linolenic, octadecatrienoic	N/A	TRFA; TRPO

TSAT = saturated fatty acid total

MUFA = monounsaturated fatty acid total

PUFA = polyunsaturated fatty acid total

TRFA = trans fatty acid total

TRMO = trans-monoenoic fatty acid total

TRPO = trans-polyenoic fatty acid total

N/A = Not available

* Totals should equal to the sum of individual fatty acids within plus or minus 10%.

Appendix D: Class of carbohydrates

Table: Class of carbohydrates

Class (Degree of Polymerization)	Sub-group	Components
Sugars (1-2)	Monosaccharides	glucose, galactose, fructose
	Disaccharides	sucrose, lactose, maltose
	Polyols	sorbitol, mannitol
Oligosaccharides (3-9)	Malto-oligosaccharides	maltodextrins
	Other oligosaccharides	raffinose, stachyose, fructo-oligosaccharides, fructans galacto-oligosaccharides
Polysaccharides (>9)	Starches	amylose, amylopectin, modified starches
	Non-starch polysaccharides	Cellulose, hemicellulose, pectins, hydrocolloids

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